

BAYESIAN HIERARCHICAL ANALYSIS OF SOCIOECONOMIC FACTORS AFFECTING LIFE EXPECTANCY IN EAST ASIAN AND PACIFIC COUNTRIES (2000-2022)

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Abstract

Life expectancy is a critical indicator of population health, yet its socioeconomic determinants vary significantly across developing countries. This study examines the impact of GDP per capita, health expenditure, clean water access, urbanisation, and fertility rates on life expectancy across 17 countries in the East Asia and Pacific (EAP) region from 2000 to 2022. Annual panel data sourced from the World Development Indicators (WDI) are analysed using Bayesian hierarchical models to account for cross-country heterogeneity. Model comparison via Leave-One-Out Cross-Validation (LOO-CV) indicates that the Bayesian hierarchical model with non-linear specifications demonstrates superior predictive performance over its linear counterpart, suggesting that the relationships between socioeconomic determinants and life expectancy are inherently non-linear across EAP countries.

1. Motivation

1.1. Background

Life expectancy at birth is one of the most widely used indicators of population health and human development. The World Health Organisation (WHO) and the United Nations Sustainable Development Goals (SDG 3) both recognise it as a key benchmark for measuring health system performance and progress toward reducing premature mortality.

The East Asia and Pacific (EAP) region presents a particularly compelling context for examining the socioeconomic determinants of life expectancy. Between 2000 and 2022, the region experienced rapid and sustained economic growth, significant expansions in public health infrastructure, and substantial reductions in poverty across many member states. However, the EAP region remains characterized by profound diversity, spanning high-income, upper middle-income and lower middle-income economies.

This diversity is totally reflected in life expectancy outcomes. As demonstrated in the exploratory data analysis (EDA) section, in 2022, Japan records an average life expectancy of approximately 84 years, compared to around 67 years in Myanmar – a gap of nearly two decades that has broadly persisted throughout the study period, notwithstanding short-term disruptions such as Cyclone Nargis in 2008 and the COVID-19 pandemic in 2020-2022. These persistent disparities suggest that economic growth alone does not fully explain differences in life expectancy, and that other socioeconomic factors may play important roles. This study therefore seeks to identify and quantify the key drivers of life expectancy across 17 EAP countries, using a modelling approach that can capture the complexity of these relationships.

1.2. Research question

This study is guided by the following research question.

Main question: What is the impact of socioeconomic factors on life expectancy across 17 EAP countries over the period 2000-2022?

To address this question, three specific sub-questions are examined:

1. Which socioeconomic factors – including GDP per capita, health expenditure access to clean water, urbanization and fertility rates – are the most significant predictors of life expectancy in the EAP region?
2. To what extent do the effects of these socioeconomic factors on life expectancy vary across countries within the EAP region?
3. Does a Bayesian hierarchical model with non-linear specifications provide superior predictive performance compared to a conventional linear model in explaining life expectancy outcomes?

Given that EAP countries share a common regional context and are therefore not fully independent and given the limited number of observations per country (23 years), a Bayesian hierarchical model is adopted as the primary analytical framework. This approach enables partial pooling of information across countries while providing full uncertainty quantification – a particularly valuable property for policy-oriented research (Gelman & Hill, 2006; McElreath, 2020).

2. The dataset

2.1. General introduction

This study draws on a panel dataset comprising 391 observations across 17 EAP countries from 2000 to 2022, sourced from the WDI. The dataset contains 11 variables, capturing both the outcome of interest and a set of socioeconomic predictors at the country-year level.

The dependent variable is *life expectancy at birth* (total, years). The independent variables include *GDP per capita* (constant 2015 US\$), *current health expenditure* (% of GDP), *access to clean drinking water* (% of population), *access to basic sanitation services* (% of population), *urban population* (% of total population), and *fertility rate* (total births per woman). Countries are further classified by income group – ranging from high-income economies to lower middle-income countries – enabling cross-group comparisons within the hierarchical modelling framework.

df_final	391 obs. of 11 variables
\$ country	: chr "Australia" "Australia" "Australia" "Australia" ...
\$ iso2c	: chr "AU" "AU" "AU" "AU" ...
\$ year	: int 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 ...
\$ income	: chr "High income" "High income" "High income" "High income" ...
\$ life_expectancy	: num 79.2 79.6 79.9 80.2 80.5 ...
..- attr(*, "label")=	chr "Life expectancy at birth, total (years)"
\$ gdp_per_capita	: num 45979 46315 47599 48510 50019 ...
..- attr(*, "label")=	chr "GDP per capita (constant 2015 US\$)"
\$ health_exp_gdp	: num 7.59 7.67 7.87 7.88 8.08 ...
..- attr(*, "label")=	chr "Current health expenditure (% of GDP)"
\$ clean_water	: num 99.7 99.7 99.7 99.7 99.8 ...
..- attr(*, "label")=	chr "People using at least basic drinking water services (% of population)"
\$ sanitation	: num 100 100 100 100 100 ...
..- attr(*, "label")=	chr "People using at least basic sanitation services (% of population)"
\$ urban_pop	: num 85.1 85.1 85.3 85.4 85.5 ...
..- attr(*, "label")=	chr "Urban population (% of total population)"
\$ fertility_rate	: num 1.76 1.74 1.76 1.75 1.77 ...
..- attr(*, "label")=	chr "Fertility rate, total (births per woman)"

2.2. Countries and panel structure

The dataset comprises 17 EAP countries observed annually from 2000 to 2022, yielding a balanced panel of 391 observations (17 countries x 23 years) with minimal missing data. Countries are classified by World Bank income group as follows:

- High income: Australia, Brunei Darussalam, Japan, New Zealand, South Korea.
- Upper middle income: China, Fiji, Indonesia, Malaysia, Mongolia, Thailand.
- Lower middle income: Cambodia, Lao PDR, Myanmar, Philippines, Samoa, Viet Nam.

This income-group classification is incorporated into the hierarchical model structure, allowing the model to partially pool information both across individual countries and across broader income groups.

2.3. Descriptive statistics.

Variable	Role	Mean	SD	Min	Median	Max
Life expectancy (years)	<i>Dependent</i>	72.73	7.19	56.36	72.98	84.56
GDP per capita (2015 US\$)	Predictor	13,442.6	—	298.6	4,232.8	61,200.5
Log(GDP per capita)	<i>Predictor</i>	8.698	1.574	5.699	8.351	11.022
Health expenditure (% GDP)	<i>Predictor</i>	4.898	1.840	1.770	4.282	12.323
Clean water access (%)	<i>Predictor</i>	88.76	12.94	47.01	94.50	100.00
Urban population (%)	<i>Predictor</i>	54.96	21.08	17.50	53.72	91.98
Fertility rate (births/woman)	<i>Predictor</i>	2.294	0.762	0.778	2.130	4.560

Table 1: Summary statistics for all variables prior to scaling.

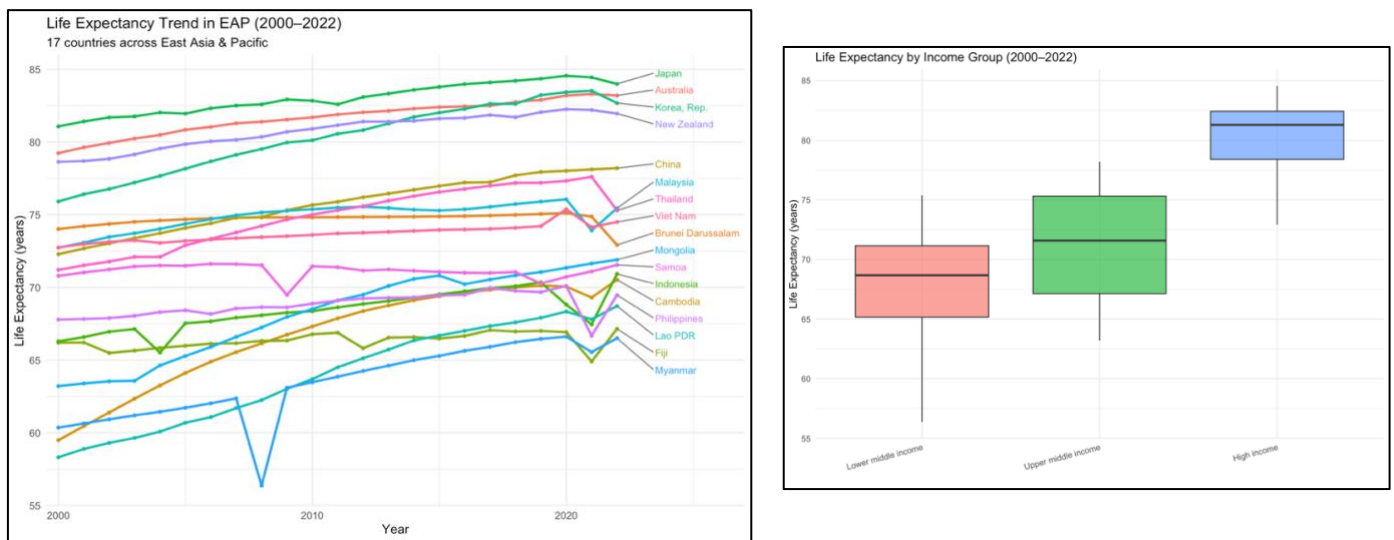
The table above presents summary statistics for all variables prior to scaling. *Life expectancy* ranges from 56.36 to 84.56 (mean = 72.73), reflecting the substantial cross-country heterogeneity. *GDP per capita* exhibits a strongly right-skewed distribution, with values ranging from \$298.6 to \$61,200.5 (constant 2015 US\$) and a mean of \$13,442.6, far exceeding the median of \$4,232.8 – which motivates the log transformation discussed in the EDA section. Other predictors also display considerable variation: *clean water access* ranges from 47.01%

to 100%, *fertility rate* from 0.78 to 4.56 births per woman, and *health expenditure* from 1.77% to 12.32% of GDP.

To ensure numerical stability and comparability across predictors, all independent variables are standardized to have mean = 0 and standard deviation = 1, the scaled dataset used for model estimation. Raw values are retained for interpretation and visualization purposes.

2.4. The exploratory data analysis (EDA)

The following figures provide an overview of the key distributional and relational features of the data that motivate the modelling choices adopted in this study.



All 17 countries show an upward trend in *life expectancy* from 2000 to 2022, though at varying rates. Two notable disruptions are visible: a clear decline in Myanmar around 2008 due to Cyclone Nargis, and a broad dip across the region in 2020-2022 attributable to COVID-19 pandemic. The boxplot confirms a clear income gradient, median *life expectancy* rises from approximately 69 years (lower middle income) to 72 years (upper middle income) and 81 years (high income), with substantially lower variability among high-income countries.

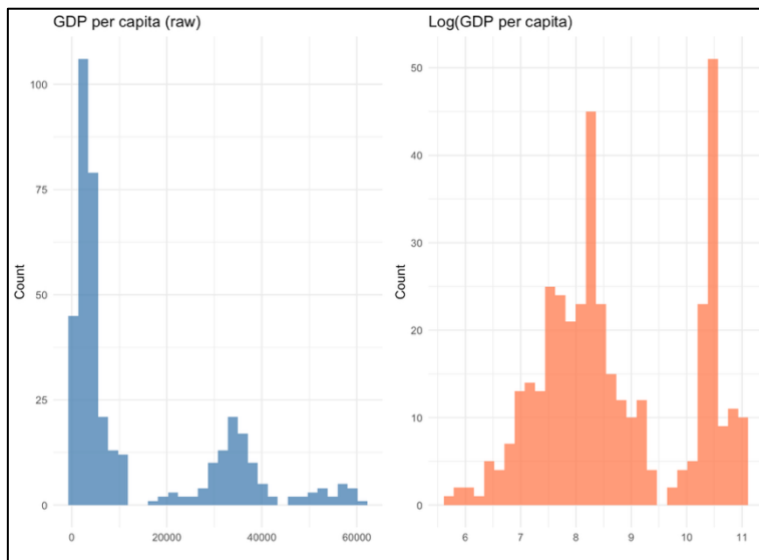


Figure 3: GDP distribution and Log transformation

In Figure 3, *GDP per capita* is heavily right-skewed, with a mean of \$13,442.6 far exceeding the median of \$4,232.8 and a small number of high-income outliers above \$60,000. Log transformation produces a near-symmetric distribution (range: 5.7-11.0), justifying its use in all model specifications.

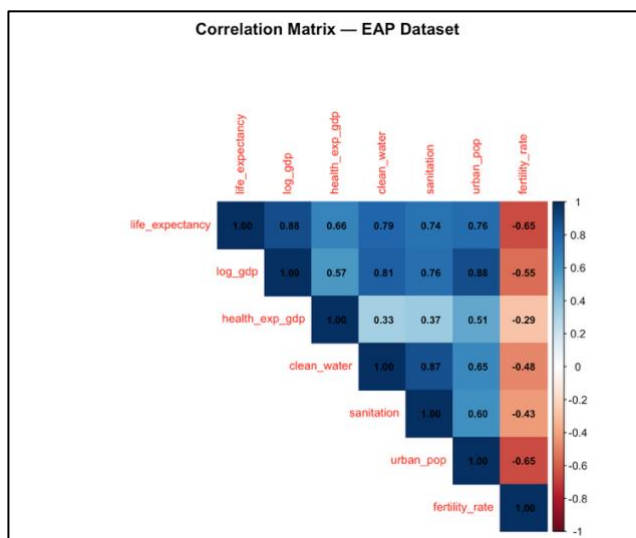


Figure 4: Collinearity & Variable Exclusion

In Figure 4, the correlation matrix identifies one critical collinearity issue: *clean water access* and *sanitation* are highly correlated ($r = 0.87$); leading to the exclusion of *sanitation* from all model specifications.

In the Figure 5 below, *Log GDP per capita* shows the strongest positive association with life expectancy ($r = 0.88$), with distinct clustering by income group. Notably, one high-income country, likely Brunei Darussalam, sits well below the trend line despite high GDP, suggesting that income alone does not fully determine health outcomes. *Clean water access* shows a strong positive association ($r = 0.79$), though the relationship flattens at high access levels, suggesting a ceiling effect; one upper middle-income country, likely Mongolia, shows high clean water access but relatively low life expectancy, further indicating that water access alone is insufficient without broader development. *Health expenditure* shows a moderate positive association ($r = 0.66$), largely driven by between-group rather than within-group differences. *Fertility rate* displays a strong and consistent negative relationship ($r = -0.65$) across all income groups.

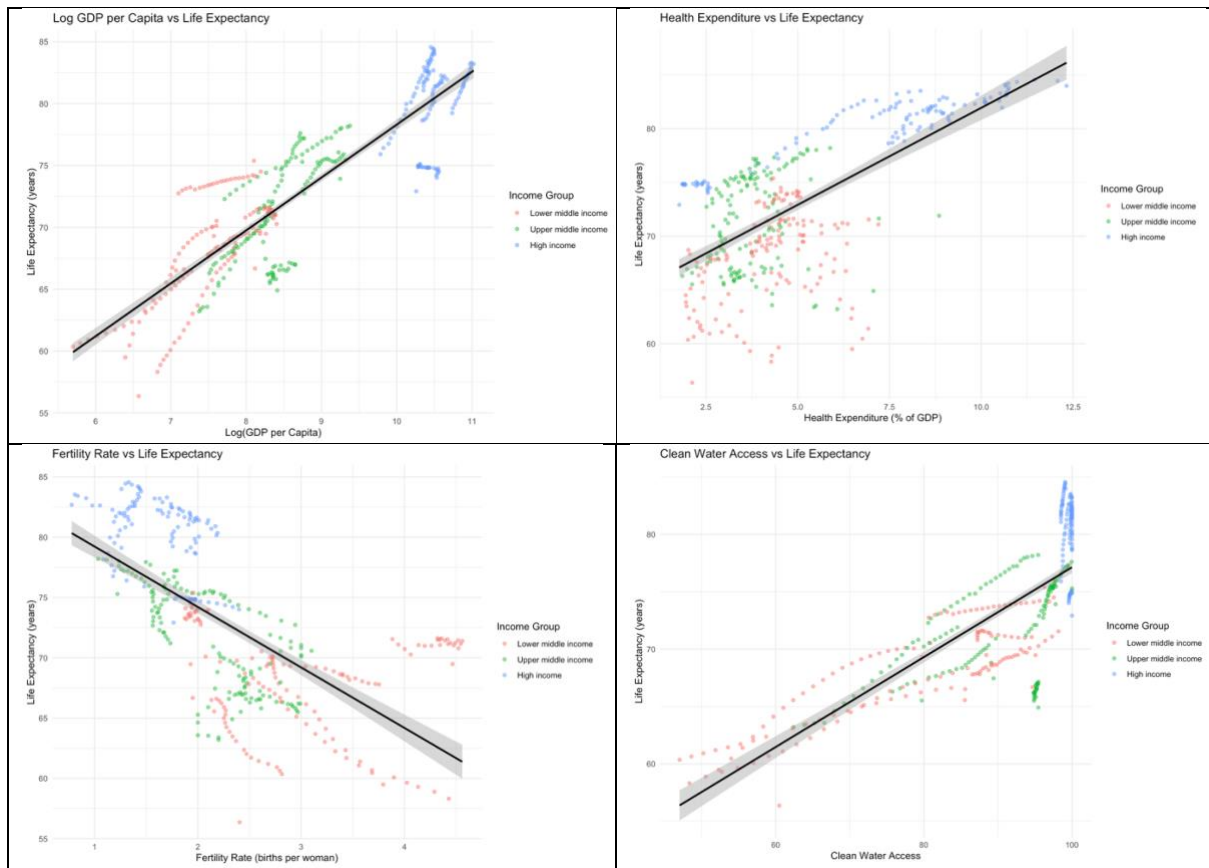


Figure 5: Bivariate scatter plots (4 panels: Log GDP, Clean Water Access, Health Expenditure, Fertility Rate)

These country-specific deviations from the overall trend suggest that unmeasured country-level characteristics play a meaningful role in determining life expectancy, providing further empirical justification for the use of a hierarchical model with country-level random effects.

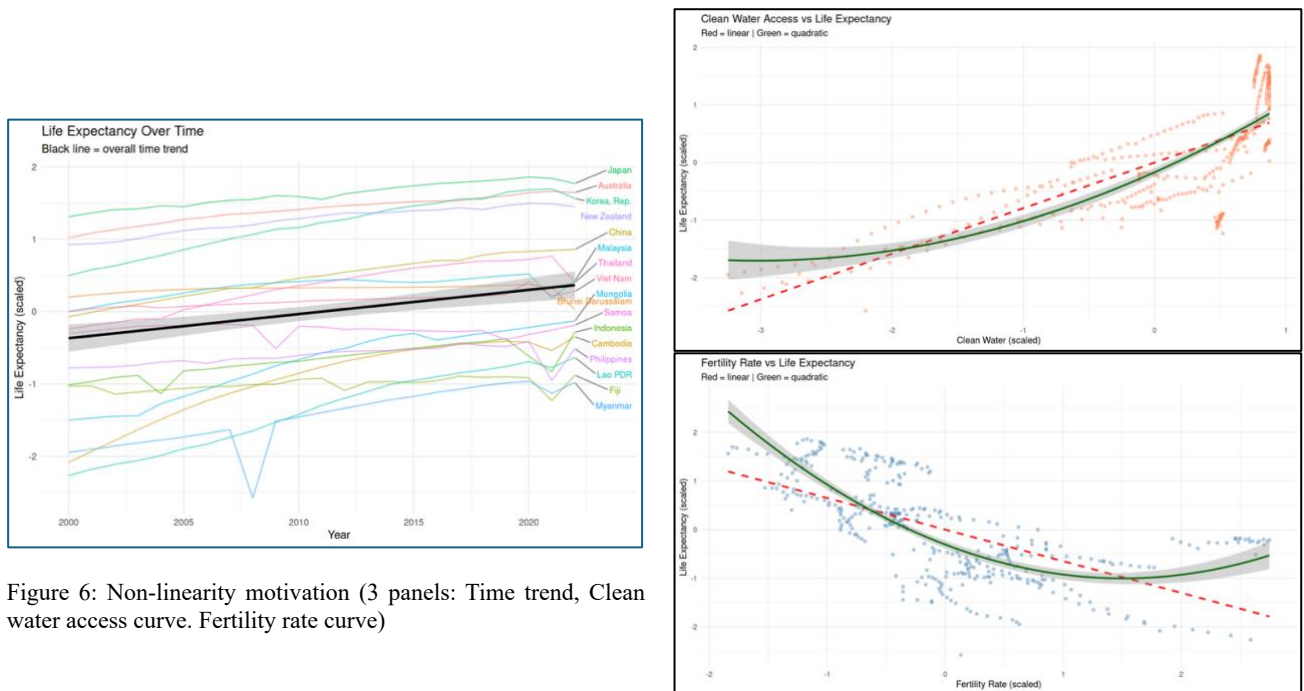


Figure 6: Non-linearity motivation (3 panels: Time trend, Clean water access curve, Fertility rate curve)

Figure 6 provides visual motivation for the non-linear model specification. The “Life Expectancy Over Time” (*time trend*) shows that while the overall regional trend is positive and approximately linear across 2000-2022, individual country trajectories deviate substantially in both level and slope – confirming that a single pooled time effect is insufficient to capture cross-country dynamics.

Clean water access illustrates why a linear specification is clearly inadequate. The quadratic fit (green) tracks the data substantially better than the linear fit (red), particularly at low levels of access where gains in life expectancy are steep, and at high levels where the relationship flattens. The linear fit systematically underestimates life expectancy at low clean water access and overestimates it at high access.

Fertility rate reveals a similarly curved relationship, the negative association is steepest at low-to-mid fertility levels but levels off at high values (scaled > 1.5), where additional increases in fertility produce diminishing reductions in life expectancy.

Taken together, these three panels provide strong empirical justification for modelling *clean water access*, *fertility rate*, and the *time trend* using flexible non-linear functions rather than linear terms.

3. Model specification

3.1. Bayesian Hierarchical Framework

Statistical inference can be approached from two inferential paradigms – frequentist and Bayesian. The frequentist paradigm treats parameters as fixed but unknown constants, drawing conclusions exclusively from observed data through point estimates and confidence intervals. The Bayesian paradigm treats parameters as random variables and quantifies uncertainty through probability distributions – combining prior beliefs with observed evidence to obtain a posterior distribution.

The foundation of Bayesian inference is Bayes’ Theorem. Given observed data $y = (y_1, \dots, y_n)$ and unknown parameters θ , the posterior is:

$$P(\theta|y) \propto L(y|\theta) \cdot \pi(\theta)$$

Where $L(y|\theta)$ is the likelihood, how probable the observed data are given θ ; and the $\pi(\theta)$ is the prior distribution. The Bayesian posterior is a full probability distribution over all plausible parameter values, providing direct and intuitive uncertainty quantification.

Why Hierarchical Structure? The 17 EAP countries are neither similar enough to be pooled together nor independent enough to be modelled separately. Countries within the same region share contextual factors – geographic proximity, trade linkages, and regional health initiatives – yet differ substantially in income level, institutional capacity, and development trajectories. A hierarchical model formalises this by treating each country's intercept as a draw from a shared regional distribution:

$$\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2), \quad i = 1, \dots, 17$$

The hyperparameters μ_α and σ_α are estimated from the data, allowing the model to automatically determine how much countries differ from one another. This partial pooling property guards against overfitting while preserving genuine cross-country differences. With only 17 countries and 23 observations per country, a fully independent (no-pooling) model would yield unstable country-level estimates. A completely pooled model, by contrast, would mask the substantial heterogeneity evident in the EAP region.

Why Non-linear Specification? A linear predictor assumes a constant marginal effect across the entire range of each predictor – an assumption that is theoretically implausible and empirically unsupported for several variables in this study.

For *clean water access* (Figure 6), the EDA shows that the quadratic fit substantially outperforms the linear fit, particularly at low access levels where initial improvements bring large health gains, and at high access levels where the relationship flattens as a ceiling effect sets in. For *fertility rate* (Figure 6), the negative association with life expectancy is steepest at low-to-moderate fertility levels but levels off at high values – consistent with demographic transition theory, where further increases in already-high fertility produce diminishing reductions in life expectancy. For the time trend (Figure 6), structural shifts across the study period – including post-MDG (Millennium Development Goals) acceleration and the sharp COVID-19 disruption in 2020-2022 – make a rigid linear time function inappropriate.

Log GDP per capita, by contrast, shows a near-linear relationship with life expectancy after log transformation (Figure 4), and is therefore retained as a linear term. Health expenditure and urban population also show no strong evidence of non-linearity in the EDA (Section 2.4) and are retained as linear terms to maintain parsimony.

To capture the non-linear relationships, penalized cubic regression splines are used for clean water access, fertility rate, and the time trend. In the Bayesian framework, the smoothing penalty acts like a prior that controls how wiggly the spline can be. The data then estimate the optimal degree of smoothing as a hyperparameter. This allows the data to determine the appropriate shape of each relationship rather than imposing a functional form a priori.

All models are implemented in R using the *brms* package (Bürkner, 2017), which uses Stan as the computational backend and the No-U-Turn Sampler (NUTS; Hoffman & Gelman, 2014) as the default MCMC algorithm.

3.2. Model specification

There are two related models. The first is a linear baseline model for comparison. The second is our final model, which incorporates non-linear terms based on the EDA findings (Section 2.4).

Linear Baseline Model

As a benchmark, we first estimate a Bayesian hierarchical model with fully linear predictors. The model is written as:

$$y_{it} = \alpha_i + \beta_1 \log_gdp_s_{it} + \beta_2 health_exp_s_{it} + \beta_3 clean_water_s_{it} + \beta_4 fertility_s_{it} + \beta_5 urban_pop_s_{it} + \varepsilon_{it}$$

$$\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \varepsilon_{it} \sim N(0, \sigma^2)$$

where: y_{it} denotes standardised life expectancy for country i in year t and α_i is a country-specific random intercept capturing unobserved country-level heterogeneity. This model serves as a benchmark against which non-linear extensions are evaluated.

Non-linear specification (Final model)

Based on the EDA findings (Section 2.4), the final model replaces the linear terms for clean water access, fertility rate and time with penalized cubic regression splines. All other predictors remain linear.

The model is specified as:

$$y_{it} = \alpha_i + \beta_1 \log_gdp_s_{it} + f_1(clean_water_s_{it}) + f_2(fertility_s_{it}) + f_3(year_s_{it}) + \beta_2 health_exp_s_{it} + \beta_3 urban_pop_s_{it} + \varepsilon_{it}$$

$$\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \varepsilon_{it} \sim N(0, \sigma^2)$$

where: f_1, f_2, f_3 denote penalized cubic regression splines (bs = "cr" in *brms* syntax).

All variables are standardised prior to model fitting. Basis dimensions are set to $k = 6$ for clean water access and fertility rate, reflecting the moderate-to-complex non-linear patterns identified in the EDA, and $k = 5$ for the time trend to ensure a smooth trajectory across the 23 annual observations while avoiding overfitting. As a general principle, larger k values allow more complex curves but risk overfitting when data are sparse; the chosen values provide sufficient flexibility while remaining well below the number of unique observations per predictor. The penalization shrinks unnecessary curvature toward linearity, so if the data do not support non-linearity, each spline effectively reduces to a linear term. Country-level heterogeneity is captured through a random intercept (1 | country), implemented in *brms* as partial pooling across the 17 EAP countries.

3.3. Prior specification

All predictors are standardised prior to model fitting. This facilitates prior specification on a common scale and ensures that the magnitudes of prior distributions are interpretable across different covariates.

The following weakly informative priors are adopted for all model parameters.

Parameter	Prior	Justification
Intercept	$N(0, 1)$	Weakly informative on standardized scale
Linear coefficients (β)	$N(0, 0.5)$	1 SD change $\rightarrow$ at most ~ 0.5 SD shift in outcome
Country-level SD (σ_α)	Exponential (1)	Mass near zero; allows larger values if needed

Residual SD (σ)	Exponential (1)	Standard weakly informative (Gelman, 2006)
Spline smoothing SD (sds)	Exponential (1)	Allows near-linearity if data do not support curvature

Table 2: Prior specifications for all model parameters

Since all predictors are standardised, $N(0, 0.5)$ for linear coefficients implies that a one-standard-deviation change in any predictor shifts life expectancy by at most approximately 0.5 standard deviation – conservative but non-restrictive. Exponential(1) priors for variance parameters place substantial mass near zero while permitting larger values if supported by the data, consistent with the weakly informative approach recommended by Gelman (2006). For spline smoothing parameters, Exponential(1) allows each smooth term to remain nearly linear if the data do not support curvature – ensuring that non-linearity is learned from the data rather than imposed by the prior.

3.4. Estimation

Models are estimated using Markov Chain Monte Carlo (MCMC) with the No-U-Turn Sampler (NUTS; Hoffman & Gelman, 2014), implemented via **brms** (Bürkner, 2017) with **Stan** as the computational backend.

For each model, four independent MCMC chains are run, each with 2,000 post-warmup draws (4,000 total iterations per chain), yielding 8,000 posterior samples. Sampler settings are configured with *adapt_delta* = 0.99 and *max_treedepth* = 12 to ensure stable posterior exploration and minimise divergent transitions. Convergence is assessed using R-hat (target < 1.01) and effective sample sizes (target > 400) (Gelman & Rubin, 1992). A fixed random seed (seed = 42) is used for reproducibility.

3.5. Model comparison – LOO cross-validation

Model selection is conducted using Leave-One-Out cross-validation (LOO-CV), which estimates out-of-sample predictive performance via the Expected Log Predictive Density (ELPD). Higher ELPD indicates better predictive performance; differences exceeding two standard errors are considered statistically meaningful.

The comparison focuses on the two models most relevant for interpretation: the linear baseline and the non-linear model. Intermediate models that were clearly outperformed are not reported.

Model	ELPD diff	SE diff	Decision
Non-linear	0.0	0.0	Selected as final model
Linear	-150.3	38.9	Rejected

Table 3: LOO-CV comparison. Δ ELPD is relative to non-linear model.

Non-linear model outperforms the linear baseline by Δ ELPD = 150.3 (SE = 38.9), approximately 3.9 standard errors. This provides strong evidence that the non-linear relationships and non-parametric time trend are genuine features of the data rather than overfitting artefacts. Therefore, the non-linear model is selected as the final model for all subsequent inference.

4. Results of the final model

4.1. MCMC Diagnostics

Convergence is assessed prior to interpreting model outputs. Trace plots for all key parameters – including the intercept, linear coefficients (log GDP per capita, health expenditure, urban population), variance parameters (σ_α , σ), and spline smoothing parameters – show stable mixing across all four chains, with no evidence of trending, drifting, or chain-specific behaviour.

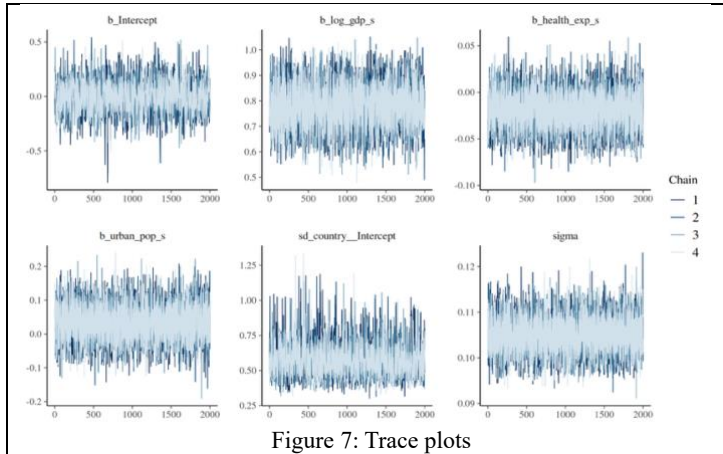


Figure 7: Trace plots

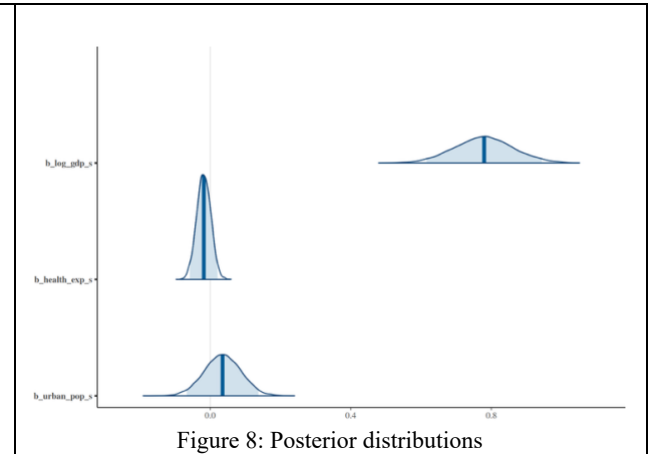


Figure 8: Posterior distributions

All parameters achieve $R\text{-hat} = 1.00$. Bulk and tail Effective Sample Sizes (ESS) are well above the recommended minimum of 400 for all parameters, confirming that the 8,000 post-warmup draws provide reliable posterior summaries. No divergent transitions are recorded, indicating that the NUTS sampler explored the posterior geometry without difficulty.

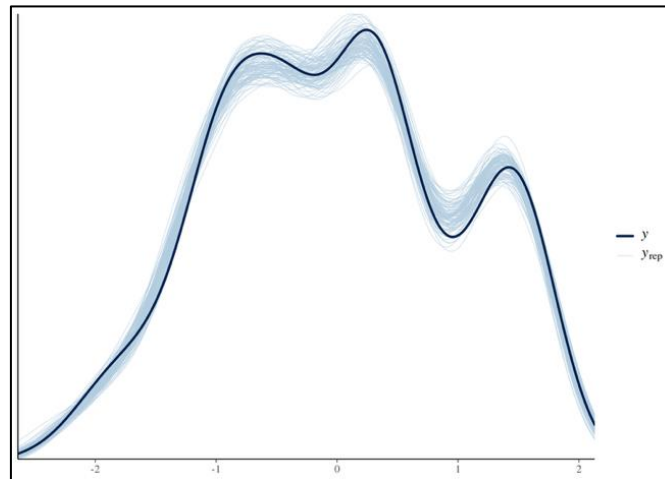


Figure 9: Posterior predictive check (pp_check)

The posterior predictive check (Figure 9) confirms acceptable model fit: the distribution of replicated datasets (y_{rep}) closely tracks the observed data (y) across the full range of life expectancy values, including both the primary peak and the secondary mode at higher values. This indicates that the non-linear model captures the key distributional features of the outcome without systematic misfit.

4.2. Posterior Estimates

```

Family: gaussian
Links: mu = identity
Formula: life_exp_s ~ log_gdp_s + s(clean_water_s, bs = "cr", k = 6) + s(fertility_s, bs = "cr", k = 6) + s(year_s, bs = "cr", k = 5) +
health_exp_s + urban_pop_s + (1 | country)
Data: df_scaled (Number of observations: 391)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

Smoothing Spline Hyperparameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sds(sclean_water_s_1)    0.11     0.07    0.05    0.28 1.00   3788   4573
sds(sfertility_s_1)      0.12     0.08    0.05    0.31 1.00   3550   5010
sds(syear_s_1)           0.05     0.05    0.01    0.16 1.00   2776   3915

Multilevel Hyperparameters:
~country (Number of levels: 17)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.57     0.11    0.40    0.84 1.00   2169   4025

Regression Coefficients:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -0.00     0.14   -0.27    0.27 1.00   1936   2716
log_gdp_s      0.78     0.08    0.62    0.94 1.00   5387   5605
health_exp_s   -0.02     0.02   -0.06    0.02 1.00   8598   6358
urban_pop_s     0.04     0.05   -0.07    0.14 1.00   8326   6036
sclean_water_s_1 -0.01     0.01   -0.03    0.01 1.00   6604   6022
sfertility_s_1  -0.09     0.02   -0.14   -0.04 1.00   8488   6277
syear_s_1       0.08     0.01    0.06    0.11 1.00   5055   5405

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma      0.11     0.00    0.10    0.11 1.00   10277   6057

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

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Figure 10: Summary output of the non-linear model

Model selection via LOO-CV is reported in Section 3.5 and the non-linear model is confirmed as the best-performing specification and is used for all subsequent inference. Figure 10 presents the posterior estimates for all regression coefficients.

Log GDP per capita is the strongest predictor of life expectancy, with a posterior mean of 0.78 (95% CI [0.62, 0.94]). The credible interval lies entirely above zero, indicating a strong and reliable positive association: a one-standard-deviation increase in log GDP per capita is associated with a 0.78-standard-deviation increase in life expectancy, conditional on other predictors. This finding is consistent with the Preston curve – higher-income countries achieve substantially longer life expectancy, with the log transformation capturing diminishing marginal returns to income at higher levels.

Health expenditure and *urban population* both show small posterior estimates with credible intervals spanning zero (health_exp_s: $\beta = -0.02$, 95% CI $[-0.06, 0.02]$; urban_pop_s: $\beta = 0.04$, 95% CI $[-0.07, 0.14]$). This indicates that, conditional on log GDP per capita, these variables contribute limited additional explanatory power. This finding is consistent with the high collinearity observed in the EDA: much of the variation in health expenditure and urbanisation is already captured by income levels.

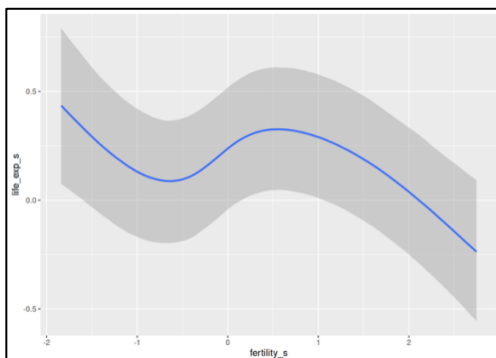


Figure 11. Conditional effect of fertility rate

Fertility rate exhibits a significant non-linear relationship, as shown in the conditional effects plot (Figure 11). The negative association is steepest at low-to-moderate fertility levels, where reductions in fertility are associated with large gains in life expectancy. At high fertility values (scaled > 1.5), the relationship levels off: further increases in fertility produce progressively smaller reductions in life expectancy. This pattern is consistent with demographic transition theory.

Time trend shows a broadly upward trajectory across the study period, with the conditional effects plot (Figure 12) revealing that the rate of improvement decelerates toward the end of the series – consistent with the regional slowdown observed during the COVID-19 period (2020–2022). This confirms that a simple linear time trend would have been insufficient to capture these dynamics.

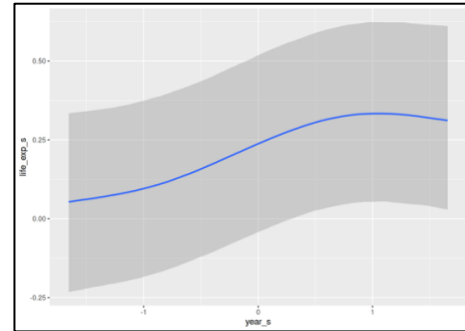


Figure 12: Conditional effect of time trend

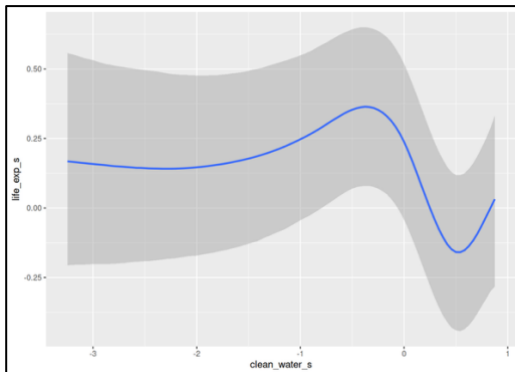


Figure 13: Conditional effect of clean water access

Clean water access is modelled with a penalized spline. The conditional effects plot (Figure 13) shows a complex pattern, with wide uncertainty at very low access levels reflecting data sparsity in that region. The 95% credible interval for the linear component of the spline spans zero, suggesting that the predictive gain from the non-linear specification – confirmed by LOO-CV – derives from the spline's flexibility in capturing local curvature rather than from a strong linear trend.

4.3. Country-level random effects

The country-level standard deviation is estimated at $\text{sd}(\text{Intercept}) = 0.57$ (95% CI [0.40, 0.84]). This is substantially larger than the residual standard deviation ($\sigma = 0.11$), confirming that **between-country variation** is the dominant source of unexplained variance in the model. This provides strong empirical justification for the hierarchical specification.

\$country				
, , Intercept				
	Estimate	Est.Error	Q2.5	Q97.5
Australia	0.2283655	0.1855626	-0.11996228	0.5984372
Brunei Darussalam	-0.6458314	0.1718400	-0.97550332	-0.3012287
Cambodia	-0.1663586	0.1591368	-0.48601436	0.1399363
China	0.4783037	0.1460173	0.19130802	0.7664656
Fiji	-0.7392155	0.1470394	-1.02743099	-0.4529231
Indonesia	-0.5342858	0.1499619	-0.83166757	-0.2501398
Japan	0.7102675	0.1745533	0.38681346	1.0573153
Korea, Rep.	0.2926546	0.1637625	-0.01731945	0.6223575
Lao PDR	-0.7518577	0.1587252	-1.07506498	-0.4505954
Malaysia	0.2874462	0.1453189	0.00416670	0.5787604
Mongolia	-0.5548276	0.1595625	-0.87130524	-0.2465535
Myanmar	-0.4353304	0.1654771	-0.76888997	-0.1231303
New Zealand	0.2786030	0.1755893	-0.05845929	0.6289343
Philippines	-0.1526038	0.1540915	-0.45511928	0.1424187
Samoa	0.3142621	0.1709910	-0.02524680	0.6444942
Thailand	0.6530938	0.1524836	0.36115555	0.9551753
Viet Nam	0.8073220	0.1528114	0.50986134	1.1036582

Figure 14: Country-level random effect from the non-linear model

Viet Nam (+0.81, 95% CI [0.51, 1.10]) and **Thailand** (+0.65, 95% CI [0.36, 0.96]) are the strongest overperformers in the region. Both are middle-income countries achieving life expectancy substantially higher than predicted by their socioeconomic indicators alone. This may reflect strong public health systems, cultural factors, or targeted health investments not captured in the model.

Japan (+0.71, 95% CI [0.39, 1.06]) also records a large positive intercept, consistent with its global reputation for longevity – driven by diet, social cohesion, and healthcare quality.

Brunei Darussalam (−0.65, 95% CI [−0.98, −0.30]) is particularly striking. Despite being a high-income country with the highest GDP per capita in the dataset, its random effect is

strongly negative – confirming the anomaly identified in the EDA. This suggests that factors beyond income, such as lifestyle, diet, or healthcare utilisation patterns, constrain life expectancy in Brunei.

Lao PDR (−0.75, 95% CI [−1.08, −0.45]) and **Fiji** (−0.74, 95% CI [−1.03, −0.45]) show the largest negative intercepts among lower middle-income countries. This indicates structural health disadvantages that persist even after accounting for their socioeconomic conditions.

Countries whose credible intervals include zero, **Australia, Cambodia, Korea, New Zealand, Philippines, and Samoa**, do not show statistically meaningful deviations from the regional mean. Their life expectancy outcomes are broadly consistent with what the model predicts given their socioeconomic profiles.

5. Conclusion

5.1. Key findings

Three main findings emerge.

First, log GDP per capita is the dominant predictor of life expectancy in the region ($\beta = 0.78$, 95% CI [0.62, 0.94]). This is consistent with the Preston curve and confirms that economic development remains the primary structural driver of population health outcomes in the EAP region.

Second, non-linear specifications substantially outperform linear alternatives. LOO-CV confirms that non-linear model – with penalized splines for clean water access, fertility rate, and the time trend – outperforms the linear baseline by a margin of $\Delta\text{ELPD} = 150.3$ (SE = 38.9). This highlights that linear models alone cannot adequately capture the relationships between socioeconomic factors and life expectancy.

Third, substantial country-level heterogeneity persists after controlling for all predictors. The country-level standard deviation is large (sd = 0.57, 95% CI [0.40, 0.84]). Notable overperformers include Viet Nam, Japan, and Thailand. Brunei Darussalam – despite its high income – records a strongly negative random effect, suggesting that factors beyond economic development play an important role in shaping health outcomes.

5.2. Limitations

Several limitations should be acknowledged.

- *Limited sample size.* The dataset covers only 17 countries, which limits generalisability of findings across the broader EAP region.
- *Aggregate-level data.* The analysis relies on country-level data, which may mask important within-country variation.
- *Observational nature.* Causal inference is not possible – the estimated associations reflect correlational patterns rather than causal effects and omitted variable bias cannot be ruled out.
- *Temporal autocorrelation.* The model assumes independent residuals, but with 23 annual observations per country, within-country residuals are likely correlated over time. While the time spline captures smooth temporal variation, unmodelled

autocorrelation may underestimate posterior uncertainty. Future work should consider an explicit correlation structure (e.g., AR(1)).

- *Random intercepts only.* The model allows country-level intercepts to vary but constrains all slopes to be identical across countries. The effect of predictors like log GDP per capita may differ between high-income and lower middle-income economies. Random slopes would address this, but the current sample of 17 countries may not support the additional complexity.

Future work could incorporate individual-level data, extend the time horizon, or explore additional predictors such as governance quality, dietary patterns, or environmental factors.

5.3. Conclusion

This project demonstrates the value of Bayesian hierarchical modelling for analysing panel data with cross-country heterogeneity and non-linear predictor relationships. The use of penalized splines within the Bayesian framework allows flexible estimation of non-linear effects while providing full uncertainty quantification through posterior distributions. The Non-linear model is selected as the final specification based on LOO-CV, confirming that both non-linear predictor relationships and non-parametric temporal dynamics are genuine features of the data. These results contribute to a better understanding of the complex, multi-dimensional drivers of life expectancy in the EAP region.

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